

PROJECT TITLE

STRATEGIC PRODUCT PLACEMENT ANALYSIS USING TABLEAU

Step 1: Problem Identification

Retail businesses generate large volumes of sales data containing information about product categories, product positions, pricing, promotions, competitor prices, consumer demographics, seasonal demand, and foot traffic. However, identifying meaningful sales patterns from raw retail datasets is difficult without proper data visualization.

This project addresses the challenge by developing an interactive Tableau dashboard that enables users to analyze retail sales performance and identify the most effective product placement strategies.

Step 2: Problem Statement

To analyze retail product placement data using Tableau and identify the key factors influencing product sales through interactive visualizations and dashboards that support data-driven business decisions-making.

Step 3: Objectives

The objectives of the project are:

- Analyze product placement strategies using Tableau.
- Identify the best-performing product categories.
- Study the impact of product positioning on sales.
- Analyze consumer demographics and purchasing behavior.
- Compare product prices with competitor prices.
- Evaluate the effectiveness of promotional campaigns.
- Analyze seasonal sales trends.
- Study the influence of customer foot traffic on sales.

- Develop an interactive Tableau Dashboard.
 - Present business insights using Tableau Story.
 - Publish the dashboard on Tableau Public.
 - Integrate the dashboard into a Flask web application.
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Step 4: Data Collection

The retail product placement dataset was collected in CSV format.

The dataset contains retail information including product categories, product positions, product prices, competitor prices, promotions, consumer demographics, seasonal demand, foot traffic, and sales volume.

Dataset Summary

- Dataset Name: Product Positioning.csv
 - Total Records: 1000
 - Total Columns: 10
 - Missing Values: None
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Step 5: Connecting Data with Tableau

The dataset was imported into Tableau using the Text File connection.

Procedure

1. Open Tableau Desktop.
 2. Select Connect → Text File.
 3. Choose Product Positioning.csv.
 4. Verify successful loading of the dataset.
 5. Confirm appropriate data types for all attributes.
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Step 6: Data Preparation

The dataset was prepared before visualization by performing the following tasks:

- Verified field data types.
- Checked for missing values.
- Confirmed dataset consistency.
- Formatted numerical values.
- Verified dimensions and measures.
- Prepared the dataset for dashboard development.

No additional calculated fields were required because the dataset already contained the required business attributes.

Step 7: Data Visualization

Eight worksheets were created to analyze different aspects of retail product placement.

Worksheet 1

Average Sales Volume vs Product Category

Displays the average sales volume across different product categories.

Worksheet 2

Consumer Demographics vs Sales Volume

Analyzes sales performance across different consumer demographic groups.

Worksheet 3

Product Category vs Price

Compares average product prices among different product categories.

Worksheet 4

Promotion Impact on Price and Sales Volume

Shows the influence of promotional campaigns on product prices and sales volume.

Worksheet 5

Average Sales Volume by Product Category by Product Position

Analyzes the impact of product positioning on sales across different categories.

Worksheet 6

Average Sales Volume by Product Category by Season

Displays seasonal sales performance across product categories.

Worksheet 7

Competitor Price vs Product Price

Compares product prices with competitor prices to evaluate pricing competitiveness.

Worksheet 8

Foot Traffic by Average Sales Volume

Shows the relationship between customer foot traffic and average sales volume.

Step 8: Dashboard Development

All eight worksheets were combined into a single interactive dashboard.

Dashboard Features

- Interactive Filters
- Bar Charts
- Pie Chart
- Donut Chart

- Bubble Chart
- Treemap
- Interactive Tables
- Responsive Layout
- Dynamic Tooltips

The dashboard enables users to explore multiple retail factors simultaneously and supports interactive business analysis.

Step 9: Story Creation

A Tableau Story was created to present the project findings in a logical sequence.

Story Points

1. Product Category Performance
2. Consumer Demographics Analysis
3. Promotion Analysis
4. Product Position Analysis
5. Foot Traffic Analysis

The Tableau Story helps users understand the relationship between different retail factors and product sales.

Step 10: Performance Testing

The dashboard was tested to ensure smooth functionality.

The following aspects were verified:

- Dataset loading performance
- Dashboard responsiveness
- Interactive filter functionality

- Visualization rendering
 - Accuracy of displayed values
 - Story navigation
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Step 11: Publishing

The completed Tableau project was published to Tableau Public.

Publishing Procedure

1. Sign in to Tableau Public.
 2. Publish the workbook.
 3. Generate the public dashboard URL.
 4. Verify online accessibility.
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Step 12: Flask Web Integration

The published Tableau Dashboard and Story were embedded into a Flask web application using the Tableau Embed API.

The Flask application provides an interactive web interface where users can explore the dashboard without opening Tableau Desktop.

Step 13: Results

The project successfully analyzed retail product placement strategies and identified important sales trends.

Major Observations

- Clothing products recorded the highest average sales volume.
- Front-of-store placement generated better sales performance.
- Promotional campaigns positively influenced product sales.

- Consumer demographics significantly affected purchasing behavior.
 - High foot traffic locations consistently achieved higher sales.
 - Seasonal demand influenced product performance.
 - Interactive dashboards simplified retail sales analysis.
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Step 14: Conclusion

The project demonstrates the effectiveness of Tableau in transforming retail sales data into meaningful business insights. Interactive dashboards and storytelling techniques enable retailers to evaluate product placement strategies, monitor sales trends, and support informed business decisions.

Step 15: Future Scope

Future improvements may include:

- Integration of real-time retail sales data.
- Sales forecasting using Machine Learning.
- AI-based product placement recommendations.
- Geographic sales analysis using maps.
- Cloud deployment for enterprise accessibility.
- Mobile-friendly dashboard implementation.